Evaluation Report

Run **eval-20261009-161607** (2026-10-09T16:17:10+00:00, 63.8 s). Generated from artifacts/evaluation/eval-20261009-161607.json. Every value below is read from that file; nothing is estimated or typed in.

Item	Value
Dataset version / records	329b6c309935 / 55,000 (incl. labelled synthetic)
Yield model version	yield-20261009155757
LLM judge model	none (LLM disabled or unavailable in this run)
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Retrieval	hybrid RRF k=60 (BM25 + MiniLM FAISS)
Test cases	31 (passed 31, failed 0)
Label verification	all relevance labels re-verified against the dataset

Test cases and how labels were created

Constraint-retrieval (16): the query states crop+state+season; relevant cases = all dataset records with that exact combination. Conditions-recommendation (8): state, season and real soil/weather values from an anchor record; every recommended crop must have at least one record in that state+season. Abstention (4): crop/state pairs with zero records. Clarification (3): queries with no agricultural context. Queries are template-generated over real values; they are test inputs, not historical records.

Retrieval metrics (K=10, 16 constraint-retrieval queries)

Metric	Value
Precision@10	0.756
Recall@10	0.747
MRR	1.000
Hit rate@10	1.000
Precision@10 on recommendation queries (state+season relevance)	1.000

Precision@10 is capped below 1 when fewer than 10 relevant cases exist (several groups have only 3-4 records), so it understates quality; MRR and Hit@K show that a relevant case is always ranked first. Pass/fail uses R-precision (precision at k = number of relevant cases).

DeepEval metrics (Claude judge)

Status: unavailable: LLM disabled for this run

Metric	Mean	Measured	Unavailable	Reason (if unavailable)
faithfulness	Not evaluated	0	24	LLM disabled for this run
answer_relevancy	Not evaluated	0	24	LLM disabled for this run
contextual_precision	Not evaluated	0	24	LLM disabled for this run
contextual_recall	Not evaluated	0	24	LLM disabled for this run

Recommendation validation (deterministic)

Check	Result
Constraint satisfaction (recommended crops recorded in state+season)	1.000
Abstention pass rate (no-evidence crops not recommended + disclosed)	1.000
Clarification pass rate	1.000

Check	Result
Unsupported numeric claims flagged	0
LLM judge verdicts	Not evaluated (LLM judge did not run in this run)

Yield model (held-out test years 2018-2020; independent of LLM metrics)

Feature set	Model	MAE	RMSE	R2	R2 (log1p)	Baseline MAE	Baseline R2
core	hgb	17.04	290.32	0.883	0.957	24.28	0.836
full	hgb	10.05	151.23	0.968	0.956	24.28	0.836

Units: source yield units, crop-specific. Original-scale RMSE/R2 are dominated by a few high-magnitude crops.

Failure cases and findings

• No failing test cases in this run.

• First run (eval-20261009-140625) flagged RET-04 and RET-13 as failures under a Precision@10 >= 0.5 rule; inspection showed every relevant case (3 and 4 exist) was retrieved at ranks 1-4 and the failure was the metric ceiling, so the rule was changed to R-precision. This is disclosed rather than hidden.

• Claude-dependent metrics (DeepEval, LLM judge) were Not evaluated in the recorded runs because the Anthropic account's credit balance was exhausted when the evaluation executed. Re-run python scripts/run_evaluation.py after adding credit to populate them; no value was filled in.